

Research Assignment Toy Megahouse

Foundations of Data Analytics and Data Science

Section A : Data fundamentals

1. What is a Database

↳ is an organized collection of information that is stored electronically so that it can be easily accessed, managed, updated and secured.

∴ Database is important in the modern world because businesses and organisations need to store and manage large amount of information efficiently.

eg. Banks : use a database to store customer details, account information and transactions.

• what is a Data warehouse

↳ a system used to collect and store large amount of data from different sources for reporting, analysis and decision making.

• A regular company might use a data warehouse to combine sales, customer, and marketing information in one place, This allows managers and analysts to create reports and make better business decisions.

eg. a Supermarket can use a data warehouse to analyse several years of sales and determine which products sell most.

What is Data Lake

• is a storage system that can store large amount of data in its original or raw form. it can contain structured, semi-structured and unstructured data.

• a business might prefer data lake when it collects very large amount of different types of data and wants to keep the original data for future analysis

• e.g. A company could store customer transactions, website logs, images, videos and JSON files in data lake

What is data mart? how does it relate to a data warehouse and who typically uses it?

• data mart is a smaller collection of data designed for a specific department or business area.

• it is often smaller part of a larger data warehouse and contains information that is relevant to a particular group

• e.g. a sales department may use a sales data mart to analyse customers, products and sales performance.

Q. What is data modeling? Why is it important when designing a system that works with data? describe 2 types of data models

• data modeling is the process of planning and organising how data will be structured, stored and connected in a system

• It is important because it helps make data easier to understand, manage and retrieve

• two types of data models

1. Conceptual data model - gives a high level view of the main data and relationships in a system

2. Logical data model - provides more detail about the data, including attributes and relationships, without focusing on how the database will physically store it.

Q. Explain the difference between structured, semi-structured and unstructured data.

• Structured data - data is organised in a fixed format usually on rows and columns (e.g. customers table)

• Semi structured data - data does not follow a strict table format but still has some organisations through labels or tags (e.g. A JSON file)

• Unstructured data - data does not have a predefined structure or format e.g. images, video or audio file.

6. What is database Management System (DBMS) name and briefly describe four popular DBMS platforms and explain a suitable use of each.

• DBMS - Software used to create, store, organise, manage and retrieve data from database

• MySQL - A relational DBMS commonly used for websites and web applications

• PostgreSQL - A relational DBMS suitable for applications that need advanced data handling and complex queries

• Oracle database - A DBMS commonly used by large organisations to manage large amount of business data.

• MongoDB - A NoSQL database that stores data in flexible document format used and is useful for applications with changing or less structured data.

7. What is data mining? how is it used in business.

• data mining is the process of analysing large amounts of data to discover useful patterns, trends and relationships

• Businesses use data mining to better understand their customers, improve decision making and identify opportunities

example

• Retail business can analyse customer purchase to identify which products are often bought together

• Banks can analyse transactions pattern to help identify unusual activity that may indicate fraud

Q. What are common data storage formats used in analytics (eg CSV, JSON, Parquet, Avro) briefly explain / describe each and state when you would use one over another.

• Common storage used in analytics include:

See

- CSV - Stores data in rows and columns. It is useful for simple datasets and spreadsheets.
- JSON - Stores data using key-value pairs. It is commonly used for APIs and exchanging data between applications.
- Parquet - Stores data efficiently in columns. It is useful for analysing large datasets because it can reduce storage and improve query performance.
- Avro - Stores data together with its structure or schema. It is useful for transferring and processing large amounts of data between systems.

• The best format depends on the type and size of data, and how the data will be used.

Q. What is ETL (extract, Transform, load) explain each step and describe why this process is important in data analytics.

• ETL > it is a process used to prepare data for analysis.

• Extract > data is collected from different sources such as database, spreadsheets or applications.

• Transform > data is cleaned, corrected and organised into useful format.

• Load > prepared data is loaded into a destination such as database or data warehouse.

- Reinforcement Learning > The model learns through trial and error by receiving rewards or penalties for its actions

11 What is artificial Intelligence (AI) define it and explain the difference between narrow AI (weak AI) and general AI (Strong AI) provide one example each

- Artificial Intelligence > is technology that allows computers and machine to perform tasks that normally require human intelligence, such as learning, understanding language and solving problems
- Narrow (Weak AI) refers to a theoretical AI that could
- Narrow AI (Weak AI) is designed to perform specific tasks eg A voice assistant that answers questions or follow commands
- General AI (Strong AI) refers to a theoretical AI that could understand, learn and perform many different intellectual tasks at a human-like level

11. What is deep learning? How does it relate to machine learning and artificial intelligence? What are neural networks and how do they work?

- Deep learning is a type of machine learning that uses neural networks and learns patterns from large amounts of data.
- Artificial intelligence is the broader field, machine learning is part of AI and deep learning is part of machine learning.
- Neural networks are computer models inspired by the way the human brain processes information, they contain connected layers that process data, learn patterns and use what they have learned to make predictions or decisions.

12. What is machine learning and how does it differ from traditional programming? Describe the entire three main types of machine learning: Supervised, unsupervised and Reinforcement learning.

- Machine learning > a type of artificial intelligence that allows computers to learn patterns from data and make predictions.
- In traditional programming, a programmer writes specific rules for the computer to follow. In machine learning the computer learns patterns ~~to follow~~ from data instead of being given every rule.
- Supervised learning > the model learns from labelled data where the correct answers are already known.
- Unsupervised learning > the model finds patterns or groups in data without being given the correct answers.

- Reinforcement Learning → The model learns through trial and error by receiving rewards or penalties for its actions

It is a sequential technique (AI) where it explores the difference between current AI (local AI) and optimal AI

(Strong AI) provide one simple task

• Artificial Intelligence → is technology that allows computers to perform tasks that normally require human

13. What is a large language model (LLM), how is it different from a standard machine learning model, name 2 well known LLMs and explain what are they used for

> large language model (LLM) is an AI model trained on a large amount of text data to understand and generate human-like language

> LLM can perform many language related tasks such as answering questions, summarising information and generating text

* GPT > used for tasks such as answering questions, writing, summarising and generating code

* Claude > used for answering questions, analysing information, writing and summarising text

14. What is a natural language processing (NLP)? describe what is it and provide 3 real world applications of NLP that you use in every day life

• Natural language processing is a field of artificial intelligence that helps computers understand, interpret and generate human language

• 3 everyday applications of NLP are

* Chatbots and virtual assistant - understand question and provide response

* Language translation - translate text from one language to another

* Email spam filters - analyse messages to identify unwanted spam emails

14. What is generative AI? how does it work at a high level? name 3 tools that use generative AI and describe what each one does.

> generative AI is a type of artificial intelligence that can create new content such as text, images, audio and computer code. It works by learning patterns from large amounts of training data and using patterns to generate new content.

3 examples of generative AI tools

- * ChatGPT > generates text and help answer questions, write content and explain information
- * Claude > Helps users generate, analyse and summarise text
- * DALL-E > Generates images based on written descriptions

15. What is a ^{large} language model (LLM) is an AI model trained on large amounts

16 What is computer vision? how do computer machines see and interpret images? provide 2 real world use cases of computer vision

> Computer vision is a field of artificial intelligence that enables computers to understand and analyse images and videos

> Machines interpret images by processing pixels and using AI models to recognise objects, Faces, Shapes and patterns

• Two real world examples

1. Facial recognition - used to recognise or verify Faces such as unlocking smartphones

2. Self driving vehicles - used to recognise roads, pedestrians, traffic signs and other vehicles

17 What is a training data ^{Set} ~~base~~? why is the quality of training data ~~base~~ data so important when building AI model

> Training dataset is a collection of data used to teach AI or machine learning model how to recognise patterns and make predictions

> The quality of training data is important because the AI model learns from the information it is given. Accurate, relevant and representative data can help model produce better results

> If the training is biased or incorrect, the model may produce inaccurate or unfair results and make poor predictions and decisions

Q. What is an API key? What is its purpose? How should it be stored and protected and what could happen if it is exposed publicly?

- An API is a unique code used to identify and authenticate a user or application when accessing an API
- Its purpose is to control and track access to the API
- > API key should be kept private and stored securely. They should not be shared publicly or included in public code repositories
- > If an API key is exposed, someone else could use it without permission, access services or data and potentially create unexpected costs or security problems

19. What is an API (Programming Interface)? Explain what it is using a simple real world analogy. Why are APIs important in modern software and data systems?

- > API (Application programming Interface) is a way for different software applications to communicate and share information with each other
- > e.g. a waiter in a restaurant, you give your order to the waiter, the waiter takes it to the kitchen and then brings the food back to you
- > An API works similarly by taking a request from the system ^(one) sending it to another system and returning the results
- > APIs are important because they allow different applications and data systems to connect, exchange data and work together.

20. What is JSON (JavaScript Object Notation) why is it commonly used when working with APIs and data exchange

• JSON is a simple text format used to store and exchange data between different systems and applications. It is commonly used with APIs because it is light weight, easy for humans to read and easy for computers to process

e.g. {
 "name": "John",
 "age": 25,
 "city": "Pretoria"
}

• This example stores information about a person using key-value pairs

21. What is a REST API? what does REST stand for and what are the main HTTP methods (GET, POST, PUT, DELETE) used in REST API's

REST API > is a way for different software applications to communicate with each other over the internet,
REST stands for Representational State transfer.

• The main HTTP methods are:

1. GET - retrieves or read data.
2. POST - Sends or create new data
3. PUT - Updates or replace existing data.
4. DELETE - Removes data.

Explain map function in python and its use

22. What is Map Function? Explain what it does in programming and data processing.

• A map Function is used to apply the same function or operation to every item in collection of data.

eg in python we can use map() to apply multiply every number in a list by 2 (2)

eg Python > result = list(map(lambda x: x * 2, numbers))
Output > 2, 4, 6, 8

23. What is prompt engineering? why is it an important skill when working with AI tools like ChatGPT or Claude?

> Prompt engineering is the process of creating clear and specific instruction for an AI system so that it can provide more useful and accurate responses

> It is important because the quality of the prompt can affect the quality of AI's response

eg poor prompt > Tell me about data

improved prompt > Explain what data analytics is and give two examples of how business use it

eg poor prompt > Write SQL

Improved prompt > Write an SQL query that calculates the total sales for each product category

20. What is a Version Control System? What is Git and why is it widely used by data analysts and developers? Explain what commit and branch are.

- > A version control system is a tool used to track and manage changes made to files and code over time.
- > Git is a popular version control system that allows data analysts and developers to track their work, collaborate with others and return to previous versions when necessary.
- > A commit is a saved record of changes made to a project.
- > A branch is a separate version of a project where changes can be made without affecting the main version.

25. What is cloud computing? Name the three main cloud service models and explain each.

- > Cloud computing is the delivery of computing services such as storage, servers, database and software over the internet.
- > 3 main cloud service models are

- IaaS (Infrastructure as a Service) - provides computing resources such as virtual servers and storage e.g. Amazon EC2.
- PaaS (Platform as a Service) - provide a platform where developers can build and run applications e.g. Google App Engine.
- SaaS (Software as a Service) - provides ready to use software that can be accessed through the internet e.g. Google Workspace.

Q. What is Python? Why is it considered one of the most popular programming languages for Data Analysis and AI?

> Python is a programming language used for data analysis, data science, machine learning and AI. It is popular because it is easy to learn, readable and has many useful libraries for working with data.

Q. Commonly used libraries in python?

- Pandas > used for cleaning, organising and analysing data
- Numpy > used for numerical calculations and working with arrays
- Matplotlib > used for creating charts and data visualisation
- Scikit-learn > used for building and testing machine-learning models

Q. What is a Jupyter Notebook? What programming language is it most commonly used with and why is it popular among data analysts and data scientists?

> A Jupyter Notebook is an interactive tool used to write and run code, add notes and display results in one place. It is most commonly used in python.

> It is popular among data analysts and data scientists because they can analyse data step by step, create visualisation, test code and explain their work in the same notebook.

3 examples of unethical use of data or AI

- Using personal data without permission → this violates people's privacy
- Biased AI Systems → this can unfairly discriminate against certain people or groups
- Storing Confidential information → this can expose private information and cause harm.

29. What is data privacy? Why is it important?

- > Data privacy is the protection and proper handling of personal information such as names, addresses, phone and numbers and financial information.
- > It is important because it protects people's personal information from being misused, stolen or accessed without permission.
- > The GDPR (general data processing regulation) → is a European Union law that protects people's personal data and gives individuals more control over how organisations use their information.

22. What is Bias in AI? Explain what it means. For an AI model to be biased provide a real world example of AI bias and describe the potential consequences.

• Bias in AI happens when an AI System produces unfair or unequal results for certain people or groups. This can happen when the data used to train the AI contains biases or does not properly represent anyone.

Example: a facial recognition system may perform better for some groups of people than others if its training data is not diverse enough.

31. What is data governance? Define it and explain the key components of a good data governance framework.

• data governance is the process of managing how an organisation collects, stores, uses and protect its data.

• key components > data quality, data security, data privacy, clear policies, defined responsibilities and access control

30. What is data ethics? provide 3 examples of unethical use of data or AI and explain what makes each example problematic.

> data ethics refers to the responsible and fair way of collecting, using, storing and sharing data.

32. What is business intelligence

Business intelligence is the process of using data to understand business performance and support better decision making

2 popular BI tools: power BI and tableau.

power BI helps business analyse data and create dashboards

Tableau helps users visualise data through charts and dashboards

33. What are the key difference between a Data Analysts, data Scientist, Data engineer and AI engineer

Career	Role	Skills	Tools
Data Analyst	Analyse data to find trends and help businesses make decisions	SQL, data cleaning Visualisation	excel, SQL power BI
Data Scientist	Uses data, statistics and Machine learning to solve complex problems and make predictions	Python, Statistics, machine learning	Python, SQL Jupyter, Notebook
Data Engineer	Builds and maintain Systems that collect store & process data.	Database, ETL, programming	SQL, Python databricks
AI engineer	develop and implement AI and machine - learning Systems	AI, Machine learning, deep learning & programming	Python Tensorflow Pytorch

32. How do you see Data and AI affecting your future career or everyday life?

• I believe Data and AI will have a big impact on my future career as I continue developing my skills in Data Analytics and I can use data to identify trends, solve business problems and support better-decision making.

eg I can use SQL, Python, power BI, data studio to analyse data and create dashboards.

• AI can help me search information, learn new skills and complete tasks more efficiently

36. Which concept did you find most difficult to understand.

• The concept I found difficult to understand was coding in SQL especially joining tables using joins. I used class notes and AI to understand it better

35. Which concept from the assignment did you find most interesting or surprising.

• It was Artificial Intelligence and machine learning. I found it interesting because machines can learn data, identify patterns and help solve real world patterns.